

PRACTICE PROGRAMS - 1

1. Program to calculate discount and bill amount using if-elif-else

In []: *#1. Program to calculate discount and bill amount using if-elif-else*

```
# Input product details
product_id = input("Enter Product ID: ")
product_name = input("Enter Product Name: ")
quantity = int(input("Enter Quantity: "))
price = float(input("Enter Price: "))

# Calculate discount based on price
if price >= 10000:
    discount = 0.50    # 50%
elif price >= 5000:
    discount = 0.30    # 30%
elif price >= 3500:
    discount = 0.15    # 15%
else:
    discount = 0.0     # No discount

# Calculate bill amount
total_price = quantity * price
discount_amount = total_price * discount
bill_amount = total_price - discount_amount

# Display all details
print("\n----- PRODUCT BILL DETAILS -----")
print(f"Product ID      : {product_id}")
print(f"Product Name     : {product_name}")
print(f"Quantity          : {quantity}")
print(f"Price              : {price}")
print(f"Discount           : {discount * 100}%")
print(f"Discount Amount    : {discount_amount}")
print(f"Bill Amount        : {bill_amount}")
```

2. Factorial of a number

In []: *#2. Factorial of a number*

```
num = int(input("Enter a number: "))
factorial = 1
if num < 0:
    print("Factorial does not exist for negative numbers.")
elif num == 0:
    print("Factorial of 0 is 1")
else:
    for i in range(1, num + 1):
        factorial *= i
    print(f"Factorial of {num} is {factorial}")
```

3. Fibonacci numbers

```
In [ ]: #3. Fibonacci numbers
ser=int(input("Enter the number of Fibonacci Series you want: "))
num1=0
num2=1
if ser<=0:
    print("Invalid Input !!!")
elif ser==1:
    print("Fibonacci Series:\n",num1)
else:
    print("Fibonacci Series:\n",num1,num2,end=" ")
    for loop in range(ser-2):
        value=num1+num2
        print(value,end=" ")
        num1=num2
        num2=value
```

4. Sum of first N natural numbers

```
In [ ]: #4. Sum of first N natural numbers
N = int(input("Enter the value of N: "))
total = 0
for i in range(1, N + 1):
    total += i
print(f"Sum of first {N} natural numbers = {total}")
```

5. Multiplication table of a number

```
In [ ]: #5. Multiplication table of a number
num = int(input("\nEnter a number to print its multiplication table: "))
print(f"\nMultiplication Table of {num}:")
for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```

6. ATM Operations - Menu Driven Program

```
In [ ]: #6. ATM Operations - Menu Driven Program
Balance=120000
ch='Y'
AcNo=int(input("Enter your Account Number: "))
Name=input("Enter your Name: ")
while ch=='Y' or ch=='y':
    print("\n\t\tWELCOME",Name)
    print("\n\t1.Deposit")
    print("\t2.Withdrawal")
    print("\t3.Balance Enquiry")
    print("\t4.Exit\n")
    choice=int(input("Choose your transaction: "))
    if choice<1 or choice>4:
        print("\n!!! CHOOSE A VALID TRANSACTION !!!\n")
    elif choice==1:
        print("\nAccount Number: ", AcNo)
        print("Name: ", Name)
        print("\t\tDEPOSIT")
        print("\t\t\t-----")
        print("Your Existing Balance is: ",Balance)
```

```

    depamt=int(input("Enter the amount to be deposited: "))
    Balance+=depamt
    print(f"\n\t\t\tAmount of Rs.{depamt} is Credited Successfully")
    print("\n\tYour Account Balance is: ",Balance)
elif choice==2:
    print("\nAccount Number: ", AcNo)
    print("Name: ", Name)
    print("\t\t\tWITHDRAWAL")
    print("\t\t\t-----")
    print("Your Existing Balance is: ",Balance)
    withamt=int(input("Enter the amount to be withdrawn: "))
    Balance=Balance-withamt
    print(f"\n\t\t\tAmount of Rs.{withamt} is Debited Successfully")
    print("\n\tYour Account Balance is: ",Balance)
elif choice==3:
    print("\n\t\tYour Account Number: ",AcNo)
    print("\t\tYour Name: ",Name)
    print("\n\t\tYour Account Balance is: ",Balance)
elif choice==4:
    break
ch=input("\nDo you want to continue transaction(Y/N)? ")

```

7. Restaurant Bill - Menu Driven Program

```

In [ ]: #7. Restaurant Bill - Menu Driven Program
choice="y"
price=0
item=''
while choice=="Y" or choice=="y":
    print('''Welcome to Star Biryani
    1. Chicken Biryani - 250
    2. Mutton Biryani - 400
    3. Fried Rice - 150
    4. Parotta - 40
    5. Exit''')

    ch=int(input("What would you like to have? (1-4): "))

    if ch==1:
        print("You have ordered Chicken Biryani")
        qty=int(input("How many you want?"))
        price=price+250*qty
        item=item+"\n"+"Chicken Biryani"
    elif ch==2:
        print("You have ordered Mutton Biryani")
        qty=int(input("How many you want?"))
        price=price+400*qty
        item=item+"\n"+"Mutton Biryani"
    elif ch==3:
        print("You have ordered Fried Rice")
        qty=int(input("How many you want?"))
        price=price+150*qty
        item=item+"\n"+"Fried Rice"
    elif ch==4:
        print("You have ordered Parotta")
        qty=int(input("How many you want?"))
        price=price+40*qty
        item=item+"\n"+"Parotta"

```

```
elif ch==5:  
    pass  
else:  
    print("Order only the available items")  
    choice=input("Do you like to have something else? (Y/N)")  
  
print("YOUR BILL",price)  
print("You had-->",item)
```